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Udemy course
Complete ML, DL, NLP
Bootcamp

Sec - 2.2

Probability Distribution
Function for Data

Lec-3 Bernoulli
Distribution

→

Lec-4 Binomial
Distribution

- The Bernoulli Distribution is the simplest discrete probability distribution.
- It represents the probability distribution of a random variable that has 2 possible outcomes: success and failure.
- It is used to model binary outcomes such as coin flip or yes/no question.

(1) Discrete Random Variables
(p.m.f)

(2) Outcomes are Binary

Example:- Tossing a coin
of H, T

$$P_n(X=H) = 0.5$$

↳ probability of
Head = 0.5

$$\begin{aligned} \text{Probability of tail} &= \\ P_n(X=T) &= 1 - 0.5 \\ &= 0.5 \end{aligned}$$

$$P_n(X=H) = 0.5 = p$$

$$P_n(X=T) = 1 - 0.5 = 0.5 = q$$

(3) Whenever the person
will pass or fail

$$P_n(X=Pass) = 0.4$$

$$P_n(X=Fail) = 1 - 0.4 = 0.6$$

(3)

⇒ Parameters for the Bernoulli Distribution include the

$$0 \leq p \leq 1$$

$$q = 1 - p$$

$K = 2 \Rightarrow 2$ outcomes

Probability of Success $\Rightarrow$
 $K = 1$

Probability of Fail $\Rightarrow$
 $K = 2$

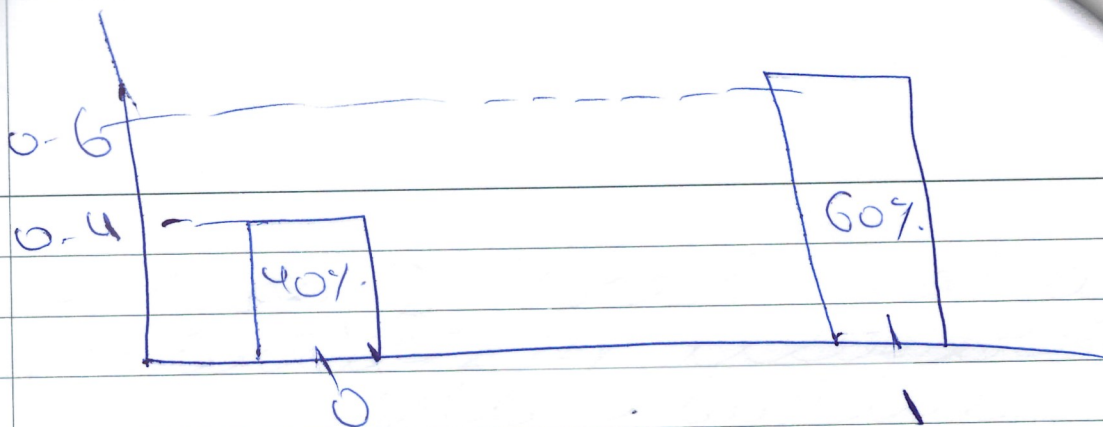
PMF (Probability mass Function)

Suppose Company has launched a new smart phone. W.

• we will use 60%.

Not used $= 100 - 60 = 40\%$.

→ the value of
 $q = 1 - p$



to construct the graph, we continue with PMF as:-

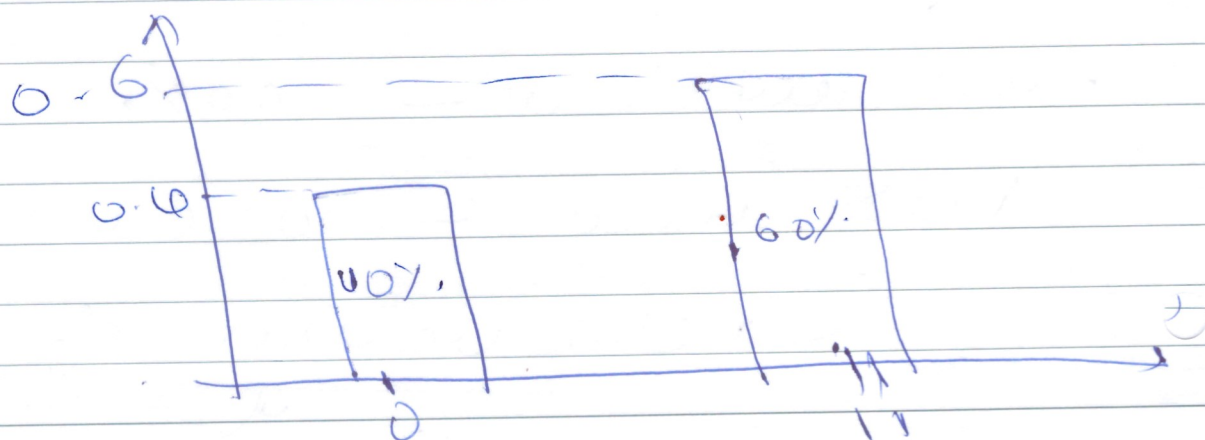
$$P_{MF} = p^k \cdot (1-p)^{1-k}$$

if $k=1$

$$P_{MF}(k=1) = p^1 (1-p)^{1-1} \\ = p^1 (1-p)^0 = p$$

$$P_{MF}(k=0) \Rightarrow p^0 (1-p)^{1-0} = (1-p) \\ = q$$

Mean of Bernoulli Distribution



(5)

⊛ Now to understand the Bernoulli distribution:-

$$E(x) = \sum_{k=0}^1 k \cdot p(k)$$

$$= 0 \times 0.40 + 1 \times 0.60$$

$$= 0 + 0.6$$

$$= 0.6$$

$$k = \{0, 1\}$$

$$p = 0.6$$

$$q = 0.4$$

$$P(1) = 0.6$$

$$P(0) = 0.4$$

$$P(1) = 0.6$$

$$P(0) = 0.4$$

⊛ Median of Bernoulli Distribution

$$\text{Median} = \begin{cases} 0 & \text{if } p < \frac{1}{2} \\ [0, 1] & \text{if } p = \frac{1}{2} \end{cases}$$

2nd median. median will be
0 if $\mu > p$

median = 0 if $\mu > p$

median = 0.5 if $\mu = p$

median = 1 if $\mu < p$

(3) Now let's talk
about Mode:-

if $p > \mu = 1$ then p will be
the mode
also μ will be the
mode

(4) Variance

$K = 0$ and 1

Probability ($K = 0$) = 0.4

Probability ($K = 1$) = 0.6 $\Rightarrow p$

$$\sigma^2 = 0.4 * (0 - 0.6)^2 + 0.6 (0.4)$$

$$\sigma^2 = 0.24 \Rightarrow$$

(7)

$$\sigma^2 = 0.24 \Rightarrow \text{probability}$$

of $K=0$ and probability
of $K=1$

$$\sigma^2 = 0.24 \Rightarrow P(K=0) + P(K=1)$$

$$\sigma^2 = 0.24$$

variance = 0.24

standard deviation, $\sigma = \sqrt{0.24}$

lec - 4

Binomial Distribution

- In the probability theory and statistics the binomial distribution is a discrete probability distribution of the number of successes in a sequence of n independent experiments.
- A single success/failure experiment is also called a Bernoulli trial or Bernoulli experiment.
- The single success experiment is called a Bernoulli trial or the Bernoulli Experiment and a sequence.
- For a single trial i.e. $n = 1$, the binomial distribution is Bernoulli distribution.

⇒ Discrete Random Variable

• Every outcome of the experiment.

• These experiments were performed for the trials.

Example:- Tossing a coin 10 times

$$n = 10$$

↓
 $\{H, T\}$

Notation = $B(n, p)$

Parameters:-

$n \in \{0, 1, 2, \dots\}$

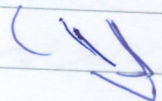


no. of trials

Notation: $B(n, p)$
 $p = T$

Parameters:

$$n \in \{0, 1, 2, \dots, \infty\}$$



no. of trials
or experiments

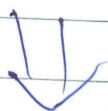
The outcome of every
experiment is 0 or 1.

Each trial belongs to:
 $p \in [0, 1] \rightarrow$
success prob. is p
probability.

$$q = 1 - p$$

Random variable:

$$K \in \{0, 1, 2, 3, \dots, \infty\}$$



Number of success

Suppose trial is a success

$\rightarrow$ No. of successes
is denoted by K

Now by applying P.M.F.:

$$P_X(k, n, p) = {}^n C_k p^k (1-p)^{n-k}$$

for $k = 0, 1, 2, \dots, n$ where

$${}^n C_k = \frac{n!}{k! (n-k)!}$$

Binomial Coefficient

We compute the mean of Binomial Distribution

$$\text{Mean} = n \cdot p$$

$$\text{Variance} = n p q$$

$$\sigma = \sqrt{n p q}$$

Standard Deviation

Ex: Coin Flip

No. of trial (n) = 5

Probability of success
(p) = 0.5

No. of success (k) = varies
from 0 to 5

Q: What is probability
of getting exactly
3 heads in 5 flips?

Soln $n = 5$, $k = 3$

$$Pr(X=3) = {}^5P_3 = \text{---}$$

Formula used:-

$$Pr(k, n, p) = {}^nC_k p^k (1-p)^{n-k}$$

$$Pr(X=3) = {}^5C_3 (0.5)^3 (1-0.5)^{5-3}$$

$$= 0.3125$$

Q.11 Example :- quality control

Scenario :- Inspecting 10 items in a factory where each item has a 10% chance of being defective

Q.12 No. of trials (n) = 10

probability of success (p) = 0.1

No. of successes (x) = varies from 0 to 10

Q.13 what is the probability of finding exactly 2 defective items in a sample of 10?

Sol.
$$Pr(X=2) = {}^{10}C_2 (0.1)^2 (1-0.1)^8$$

$= 0.1937 \checkmark$